KAIZEN MATHS

Lesson starter

Differential Equations: First and Second Order

First-Order Separable

Name

S1

Solve each separable differential equation.

$\frac{dy}{dx} = 4xy$

S2

Solve each separable differential equation and use the initial condition.

$\frac{dy}{dx} = 4xy$, with $y = 2$ when $x = 0$

S3

Solve each exponential growth or decay differential equation.

$\frac{dy}{dx} = -3y$, with $y = 8$ when $x = 0$

S4

Solve each separable differential equation.

$\frac{dy}{dx} = 5xy$

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KAIZEN MATHS

Exit ticket

Differential Equations: First and Second Order

First-Order Separable

Name _____

E1

Solve each separable differential equation and use the initial condition.

dy dx = 4x y , with y = 4 when x = 0

E2

Solve each exponential growth or decay differential equation.

dy dx = 3y, with y = 10 when x = 0

E3

Solve each separable differential equation.

$\frac{dy}{dx} = 5x^3 y$

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Solve each exponential growth or decay differential equation.

dy dx = 3y, with y = 10 when x = 0

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Solve each separable differential equation.

$\frac{dy}{dx} = 5x^3 y$

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Differential Equations: First and Second Order

First-Order Separable

Name _____

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Solve each separable differential equation and use the initial condition.

dy dx = 4x y , with y = 4 when x = 0

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Solve each exponential growth or decay differential equation.

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E3

Solve each separable differential equation.

$\frac{dy}{dx} = 5x^3 y$

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Teacher answers & working

Differential Equations: First and Second Order · First-Order Separable

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S1

Solve each separable differential equation.

$dy\,dx = 4x\,y$

Answer:

$y^2 = 4x^2 + C$

Working:

Multiply both sides by y and by dx.

$y\,dy = 4x\,dx$

Integrate both sides.

$\int y\,dy = \int 4x\,dx$

$\frac{y^2}{2} = \frac{4x^2}{2} + C$

Multiply through by 2.

$y^2 = 4x^2 + C$

S2

Solve each separable differential equation and use the initial condition.

dy dx = 4x y , with y = 2 when x = 0

Answer:

$y^2 = 4x^2 + 4$

Working:

Multiply both sides by y and by dx.

$y\,dy = 4x\,dx$

Integrate both sides.

$\frac{y^2}{2} = \frac{4x^2}{2} + C$

Multiply through by 2.

$y^2 = 4x^2 + C$

Use y = 2 when x = 0.

$2^2 = 4(0)^2 + C$

$C = 4$

$y^2 = 4x^2 + 4$

Teacher answers & working

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S3

Solve each exponential growth or decay differential equation.

dy dx = -3y, with y = 8 when x = 0

Answer:

$y = 8e^{-3x}$

Working:

Separate the variables.

$\frac{1}{y}\,dy = -3\,dx$

Integrate both sides.

$\int \frac{1}{y}\,dy = \int -3\,dx$

$\ln|y| = -3x + C$

Write the solution in exponential form.

$y = Ae^{-3x}$

Use y = 8 when x = 0.

$8 = Ae^{0}$

$A = 8$

$y = 8e^{-3x}$

S4

Solve each separable differential equation.

$dy\,dx = 5x\,y$

Answer:

$y^2 = 5x^2 + C$

Working:

Multiply both sides by y and by dx.

$y\,dy = 5x\,dx$

Integrate both sides.

$\int y\,dy = \int 5x\,dx$

$\frac{y^2}{2} = \frac{5x^2}{2} + C$

Multiply through by 2.

$y^2 = 5x^2 + C$

Teacher answers & working

Differential Equations: First and Second Order · First-Order Separable

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E1

Solve each separable differential equation and use the initial condition.

dy dx = 4x y , with y = 4 when x = 0

Answer:

$y^2 = 4x^2 + 16$

Working:

Multiply both sides by y and by dx.

$y\,dy = 4x\,dx$

Integrate both sides.

$\frac{y^2}{2} = \frac{4x^2}{2} + C$

Multiply through by 2.

$y^2 = 4x^2 + C$

Use y = 4 when x = 0.

$4^2 = 4(0)^2 + C$

$C = 16$

$y^2 = 4x^2 + 16$

E2

Solve each exponential growth or decay differential equation.

dy dx = 3y, with y = 10 when x = 0

Answer:

$y = 10e^{3x}$

Working:

Separate the variables.

$\frac{1}{y}\,dy = 3\,dx$

Integrate both sides.

$\int \frac{1}{y}\,dy = \int 3\,dx$

$\ln|y| = 3x + C$

Write the solution in exponential form.

$y = Ae^{3x}$

Use y = 10 when x = 0.

$10 = Ae^0$

$A = 10$

$y = 10e^{3x}$

Teacher answers & working

Differential Equations: First and Second Order · First-Order Separable

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E3

Solve each separable differential equation.

$dy\,dx = 5x^3\,y$

Answer:

$y^2 = \frac{5}{2}x^4 + C$

Working:

Multiply both sides by y and by dx.

$y\,dy = 5x^3\,dx$

Integrate both sides.

$\int y\,dy = \int 5x^3\,dx$

$\frac{y^2}{2} = \frac{5x^4}{4} + C$

Multiply through by 2.

$y^2 = \frac{5}{2}x^4 + C$